NNN-R-0055(DM)
23 January 1968

INTERIM REVISION OF
Fed. Spec.GG-S-803a
11 June 1952 and
Fed. Spec. LLL-R-53
11 April 1945

INTERIM FEDERAL SPECIFICATION

RACK, TEST TUBE, LABORATORY

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-S-803a dated 11 June 1952, and LLL-R-53, dated 11 April 1945.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metal laboratory test tube racks.

1.2 Classification. Racks covered by this specification shall be of the following types and sizes as specified (see 6.2).

Type I
Size 1. 14 tube (see figure 1).

Type II
Size 2. 19 tube (see figure 2).

Type III
Size 3. 40 tube (plastisol coated).

FSC 6640

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 2240-64T	Method of Test for Indentation Hardness of Rubber and Plastics by Means of a Durometer (Tentative).
D 638-64T	Method of Test for Tensile Properties of Plastics (Tentative).
D 624-54	Methods of Test for Tear Resistance of Vulcanized Rubber.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality, best suited for the purpose intended, and as hereinafter specified.

3.1.2 Corrosion-Resisting Steel. Racks shall be constructed of No. 24 U.S. standard gage (0.025 inch thick) corrosion- resisting steel conforming with Fed Std 66, AISI 304.

3.2 Design and construction

3.2.1 Type I.

3.2.1.1 Size 1. Size 1 rack shall be a double shelf and base, double-row test-tube support capable of holding fourteen 32 millimeter (mm.) diameter test tubes. Style and design shall be in strict accordance with figure 1. Size 1 test-tube rack shall consist of two 14-hole shelves (top and middle decks) and an unperforated base or bottom deck. Each of the fourteen holes in the top shelf shall be in vertical alinement with a similar hole in center shelf. Back row of holes in each shelf shall be offset from front row to make all test tubes visible from the front. Each hole shall be 1-11/32 inches (34 mm.) in diameter, minus tolerance 0, plus tolerance 1/32 inch. The support shall measure 15 inches long by 4-3/16 inches wide by 4-3/16 inches high and shall be made of three pieces as illustrated. Top and ends shall be one-piece sheet with neatly mitred corners. Free edges of top and ends, except at base, shall be flanged 90° for a distance of approximately 7/16 inch with a suitable 180° return. Free edges-(long dimensions) of middle shelf and unperforated bottom shall be similarly treated. Each of the two short dimensions of the middle shelf and the unperforated bottom shall be flanged 90° for a distance of approximately 7/16 inch, without the 180° return, to present a surface for attachment to the ends of the rack. The middle shelf and the base shall be set inside the flanged ends of the rack and secured by four spot welds on each end as indicated in figure 1. A flat triangle of the same material as the rack proper shall be fitted inside of each of the four top corners of the rack

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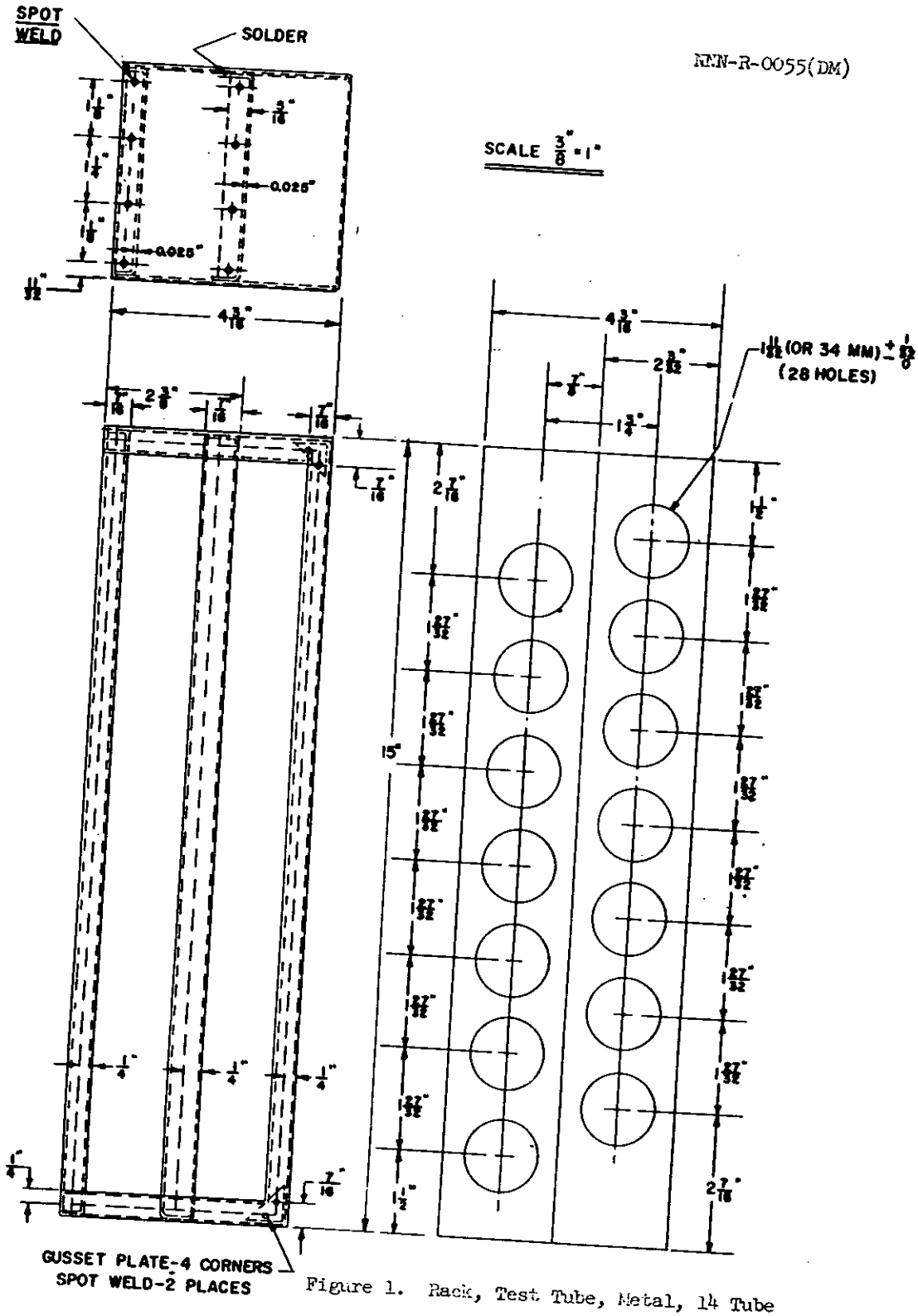


Figure 1. Rack, Test Tube, Metal, 14 Tube

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across the mitre, and shall lie flat against, and join the flange of the end and top in each of the corners. Triangles shall be affixed by two spot welds each, and shall serve to increase the rigidity of the rack. Flanges of the inset middle shelf and bottom shall be soldered to the flange of the ends to finish off the appearance of the inset. Solder shall be ground flat and smooth.

3.2.2 Type II.

3.2.2.1 Size 2. Size 2 shall be a circular test tube rack suitable for use in water baths, and shall be capable of holding nineteen 13-mm. diameter test tubes. Style and design shall be in strict accordance with figure 2. Support shall be approximately 5-5/8 inches in diameter and approximately 7-3/4 inches in over-all height. The rack shall consist of three circular shelves, each shelf flanged 90° downward for a distance of approximately 1/8 inch around the circumference. Shelves shall be approximately 1-1/2 inches apart and shall be supported in position by four vertical 1/8 inch-diameter uprights evenly spaced around the circumference just inside the flange of the shelves. Upright supports shall be fitted through the three shelves and shall be soldered into position on bottom surface of each shelf so that approximately 3/4 inch continues through the bottom shelf to act as four legs. Top of each of the four supports shall be cut flush with top of the upper shelf and polished smooth. A central stem shall be fitted through the center of the shelves. The bottom of the central stem shall also be carried through the bottom shelf to act as a fifth leg and the top of the stem shall be continued upward for the balance of the overall height and shall terminate in a ring of approximately 1-inch inside diameter to act as a lift ring. Center stem shall also be soldered in position on bottom surface of each shelf.

3.2.2.1.1 Alternate. As an alternate, assembly of shelves may also be accomplished by fitting the rods through the shelves which are held in position by tube spacers between the shelves and between the top shelf and the lift ring. Rods shall be peened or spun over at both ends except the center rod which terminates in the lift ring at the top. Tube spacers shall be corrosion-resisting steel tubing. Middle and top shelves shall each have 19 holes properly spaced in two rings on each shelf, in such a manner that each hole on the middle shelf lines up vertically with the hole in top shelf so that 19 test tubes can be fitted into the holes. Holes shall be 1/16 inch in diameter, minus tolerance zero, plus tolerance 1/32 inch. Bottom shelf shall be perforated with not less than 24 holes (each approximately 1/8 inch diameter) to act as drains for water when support is lifted out of bath. Holes in bottom shelves shall be located so that they will not be covered by test tube bottoms when tubes are in position in support. Upright supports and center stem shall be approximately 1/8-inch diameter corrosion-resisting steel wire. Solder shall be ground flat and smooth.

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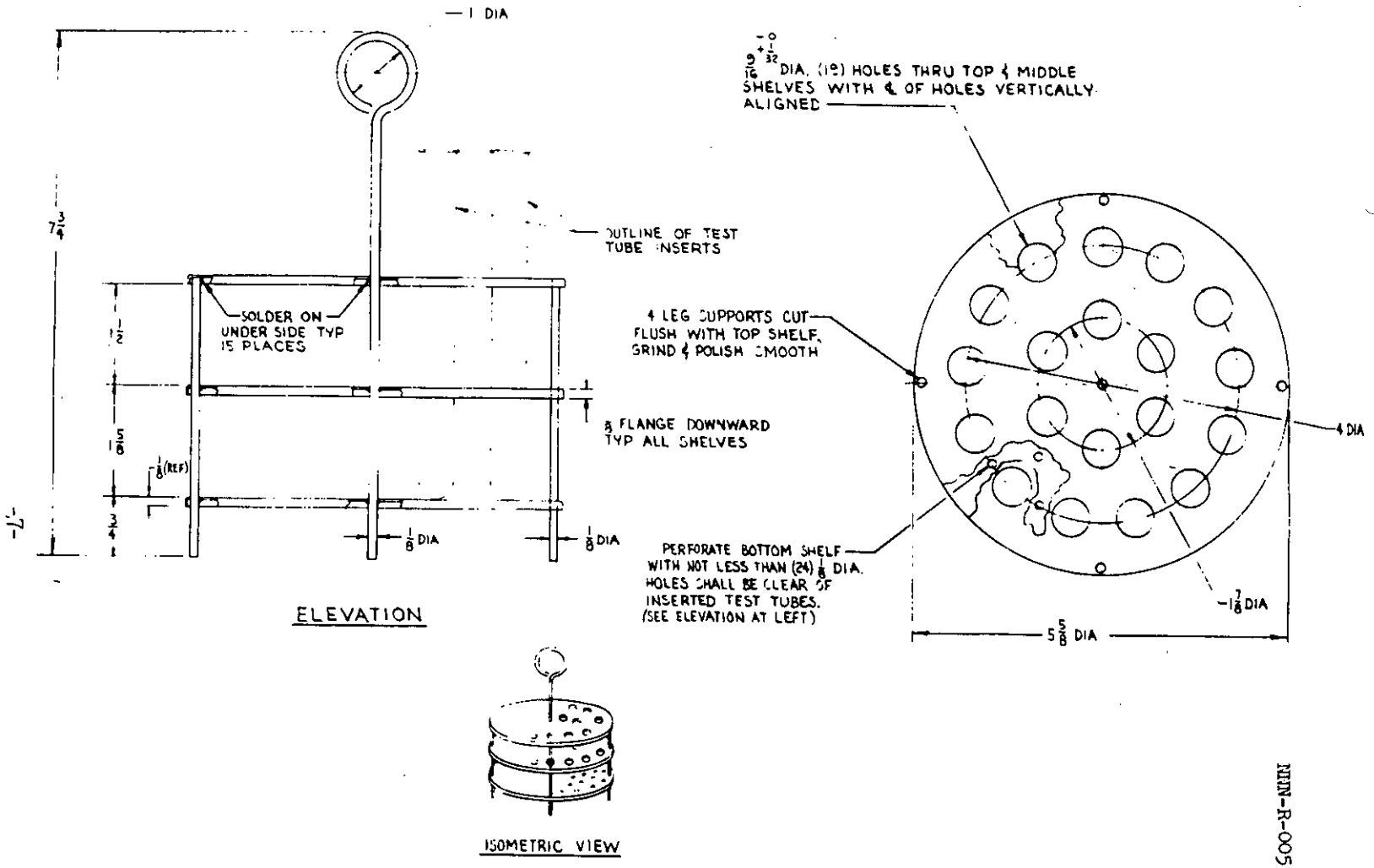


Figure 2. Rack, Test Tube, Metal, Circular, 19 Tube

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3.2.3 Type III. The test tube support rack shall be a steel wire assembly 4 inches high by 4 inches wide by 10 inches long (4 openings wide by 10 openings long). Dimensional tolerances shall be ± 1/16 inches. Assembly shall be made up on three wire frames, spaced vertically. The lower frame shall be expanded metal mesh to support the bottom of the tubes. Each upper frame shall consist of spaced wires to provide 40 alined openings 7/8 inch ± 1/32 inch square, capable of securely holding test tubes 20 mm. in diameter in a vertical position. Construction shall be resistance welded throughout. This assembly shall be completely coated with a durable vinyl plastisol (see table I).

Table I

Physical Properties		ASTM Method No.
Specific Gravity at 80°	1.30 ± 0.02	
Durometer "A" Hardness	80 ± 2	D-2240-64T
Tensile Strength	1900 p.s.i. (min.)	D-638-64T
Tear Strength	275 lbs. inch(min.)	D-624-54
Maximum Temperature Limit	250°F (see 4.4.1)	
Flammability	(see 4.4.2)	
Fusion to Primed Metal	(see 4.4.3)	
Plasticizer Components	(see 4.4.4)	

3.3 Finish for Types I and II. All surfaces of the racks shall be given a No. 4 commercial finish.

3.4 Workmanship. Workmanship shall be first class throughout, and shall insure a sturdy, rigidly constructed rack. Racks shall be free from defects which detract from their appearance or which might impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

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4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Max. Temp. Limit	3.2.3	4.4.1	S-1	*
Flammability	3.2.3	4.4.2	S-1	*
Fusion to primed metal	3.2.3	4.4.3	S-1	*
Plasticizer Components	3.2.3	4.4.4	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The racks shall be examined to determine compliance with all requirements contained in this specification.

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4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *.

Major
101 Rack not free from sharp edges.
102 Rack not free from deformation.
103 Soldering not secure.
104 Holes on shelves not vertically alined.
105 Rack not completely coated.

Minor
201 Holes on shelves not round.
202 Racks not free from dirt, grease, or foreign particles.

* Examination shall not be restricted to the defects listed above.

4.3.2 Dimensional examination. The racks shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Maximum Temperature Limit. Sample shall be placed in oven at 250° F and allowed to stand 30 minutes. Plastisol shall not distort or become tacky.

4.4.2 Flammability. Plastisol film shall be immediately self extinguishing after 30 second exposure to bunsen burner ignition.

4.4.3 Fusion to primed metal. Pull required to separate a 1-inch wide coating strip shall be determined by the deadweight method.

4.4.4 Plasticizer components. Shall resist hydrolysis when refluxed at the boiling point in 10 percent sulfuric acid for a minimum of 100 hours.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One rack, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

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5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military agencies.

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5.3.2.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test tube racks covered by this specification are intended to be used to support test tubes during routine laboratory procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

NNN-R-0055(DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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